Machine Learning — Quick Reference

1. What is Machine Learning?

Machine Learning (ML) is a branch of artificial intelligence in which computers learn patterns from data and use those patterns to make predictions or decisions. Instead of explicitly programming every rule, we provide examples and allow a model to learn from them.

2. Supervised Learning

Supervised learning uses labeled data. Each training example contains an input and a known target output. The model learns a relationship between inputs and outputs. Common tasks include classification and regression.

3. Unsupervised Learning

Unsupervised learning works with data that does not have labeled target values. The goal is to discover useful structure or patterns in the data. Clustering and dimensionality reduction are common examples.

4. Classification

Classification predicts a discrete category or class. Examples include predicting whether an email is spam or not spam, and classifying an image as a cat or dog.

5. Regression

Regression predicts a continuous numerical value. Examples include predicting house prices, temperature, or sales revenue.

6. Training and Testing Data

A dataset is commonly divided into training and testing sets. The training set is used to learn model parameters, while the testing set is used to evaluate how well the trained model generalizes to unseen data.

7. Overfitting

Overfitting occurs when a model learns the training data too closely, including noise or accidental patterns. Such a model may perform very well on training data but poorly on unseen data.

8. Underfitting

Underfitting occurs when a model is too simple to capture important patterns in the data. It generally performs poorly on both training and unseen data.

9. Bias–Variance Trade-off

Bias is error caused by overly simple assumptions in a model. Variance is error caused by excessive sensitivity to the training data. A useful model balances bias and variance so that it generalizes well.

10. Common Machine Learning Algorithms

Linear Regression is commonly used for regression. Logistic Regression, Decision Trees, Random Forests, and Support Vector Machines can be used for classification. K-Means is a popular unsupervised clustering algorithm.

11. Model Evaluation

For classification, common metrics include accuracy, precision, recall, and F1-score. For regression, common metrics include mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE).

12. Example

Suppose a model is given historical house data containing area, number of rooms, and location along with known prices. A supervised regression model can learn from these examples and then predict the price of a new house.

Practice Questions

1. What is the main idea behind Machine Learning?

2. What is the difference between supervised and unsupervised learning?

3. What is the difference between classification and regression?

4. What is overfitting?

5. What does the bias–variance trade-off describe?

6. Name two classification algorithms.

7. Name one common clustering algorithm.

8. Why do we use a test set?